

Data Analysis Presentation

BCA/BTech Degree Efficiency vs Student Personality Fit

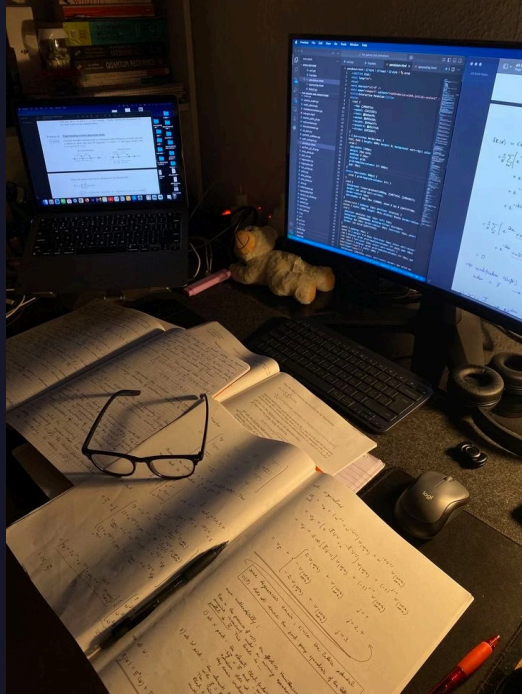
Data Preparation & Exploratory Data Analysis

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Introduction & Background

The Core Question

Is a technical degree still effective — and does it match who you are as a learner?



01

Online Learning Boom

Rapid growth of online learning platforms has raised questions about traditional degree relevance

02

Student Reality

Students face stress, confusion, and mismatch between their personality and course demands

03

Study Focus

Technical education requires logical thinking — not all students adapt to the same teaching methods; psychological factors influence outcomes

Problem Statement & Objectives



The Problem

Students struggle with technical subjects due to mismatch between personality and course structure, lack of practical exposure, and inadequate guidance. The effectiveness of current education systems needs evaluation.



Study Objectives

- Analyze student learning behavior and personality impact on performance
- Identify stress factors in technical education
- Evaluate employability confidence and suggest curriculum improvements



Psychological Framework

Learning is influenced by personality traits — two models were applied



Openness

Curiosity & innovation



Conscientiousness

Discipline & consistency



Extraversion

Social & group learning



Agreeableness

Cooperation



Neuroticism

Anxiety & stress



MBTI Types

Analysts, Diplomats, Sentinels, Explorers



Understanding the Big Five (OCEAN)

These five broad personality traits profoundly influence how students engage with, adapt to, and succeed in technical education environments.



Openness to Experience

Curiosity, imagination, and a willingness to explore new concepts. Crucial for innovation and adapting to evolving technologies.



Conscientiousness

Self-discipline, organization, and goal-directed behavior. Key for managing complex coursework and project deadlines.



Extraversion

Sociability and assertiveness. Impacts participation in group projects, networking, and seeking support from peers and instructors.



Agreeableness

Cooperation, empathy, and compassion. Facilitates teamwork and effective communication in collaborative learning settings.



Neuroticism

Tendency towards emotional instability, anxiety, or moodiness. High levels can hinder academic performance and well-being under pressure.

The Big 5 Personality Traits





Architect
INTJ-A / INTJ-T

Imaginative and strategic thinkers, with a plan for everything.



Logician
INTP-A / INTP-T

Innovative inventors with an unquenchable thirst for knowledge.



Commander
ENTJ-A / ENTJ-T

Bold, imaginative and strong-willed leaders, always finding a way – or making one.



Debater
ENTP-A / ENTP-T

Smart and curious thinkers who cannot resist an intellectual challenge.

Diplomats



Advocate
INFJ-A / INFJ-T

Quiet and mystical, yet very inspiring and tireless idealists.



Mediator
INFP-A / INFP-T

Poetic, kind and altruistic people, always eager to help a good cause.



Protagonist
ENFJ-A / ENFJ-T

Charismatic and inspiring leaders, able to mesmerize their listeners.



Campaigner
ENFP-A / ENFP-T

Enthusiastic, creative and sociable free spirits, who can always find a reason to smile.

Sentinels



Logistician
ISTJ-A / ISTJ-T

Practical and fact-minded individuals, whose reliability cannot be doubted.



Defender
ISFJ-A / ISFJ-T

Very dedicated and warm protectors, always ready to defend their loved ones.



Executive
ESTJ-A / ESTJ-T

Excellent administrators, unsurpassed at managing things – or people.



Consul
ESFJ-A / ESFJ-T

Extraordinarily caring, social and popular people, always eager to help.

Explorers



Virtuoso



Adventurer



Entrepreneur



Entertainer

PERSONALITY DEEP DIVE

MBTI Archetypes: Understanding Behavioral Tendencies

The Myers-Briggs Type Indicator (MBTI) categorizes individuals into 16 distinct personality types, grouped into four overarching archetypes. Understanding these provides deeper insights into student preferences for learning, collaboration, and problem-solving.



Analysts

Driven by logic and innovation, these types (INTJ, INTP, ENTJ, ENTP) excel in strategic thinking and complex problem-solving.



Diplomats

Empathetic and cooperative (INFJ, INFP, ENFJ, ENFP), they thrive in harmonious environments, seeking understanding and connection.



Sentinels

Practical and orderly (ISTJ, ISFJ, ESTJ, ESFJ), Sentinels value stability, responsibility, and adherence to established methods.



Explorers

Adventurous and spontaneous (ISTP, ISFP, ESTP, ESFP), they prefer hands-on experience and adapting to new situations.

Our Data Analysis Workflow

To ensure robust and reliable insights from our student dataset, we followed a structured approach to data analysis, encompassing several critical stages.



This systematic workflow ensures that raw data is meticulously transformed and examined, laying the groundwork for a deeper understanding of student personality and academic outcomes.

Dataset & Data Preparation

Dataset Description

Source: Google Form Survey ·
Sample: ~80 students · Age:
18–23 · Questions: 30+ · Data
Type: Quantitative +
Qualitative

Cleaning & Imputation

Missing values handled using
mean imputation. Data cleaned
and standardized for
consistency across all
variables.

Encoding & Text Mining

Categorical data converted to numeric using Likert scale. Open-ended responses processed using text mining techniques.

Dataset Description (Tabular Form)

Attribute	Description
Dataset Title	BCA/BTech Degree Efficiency vs Student Personality Fit
Data Source	Primary data collected through Google Form survey
Target Population	Undergraduate students (BCA/BTech)
Sample Size	Approximately 80 respondents
Age Group	18–23 years
Academic Year Distribution	Mostly 2nd year students (with some from 1st, 3rd, 4th year)
Type of Data	Mixed (Quantitative + Qualitative)
Data Collection Method	Online questionnaire
Response Type	Likert Scale (1–5), Multiple Choice, Open-ended

Exploratory Data Analysis

Key findings from analyzing the survey data across demographics, curiosity, and study habits.

Demographics

Majority aged 18–23, mostly 2nd-year students with similar academic backgrounds

Study Schedule

Regular study habits improve confidence; irregular schedules lead to lower performance

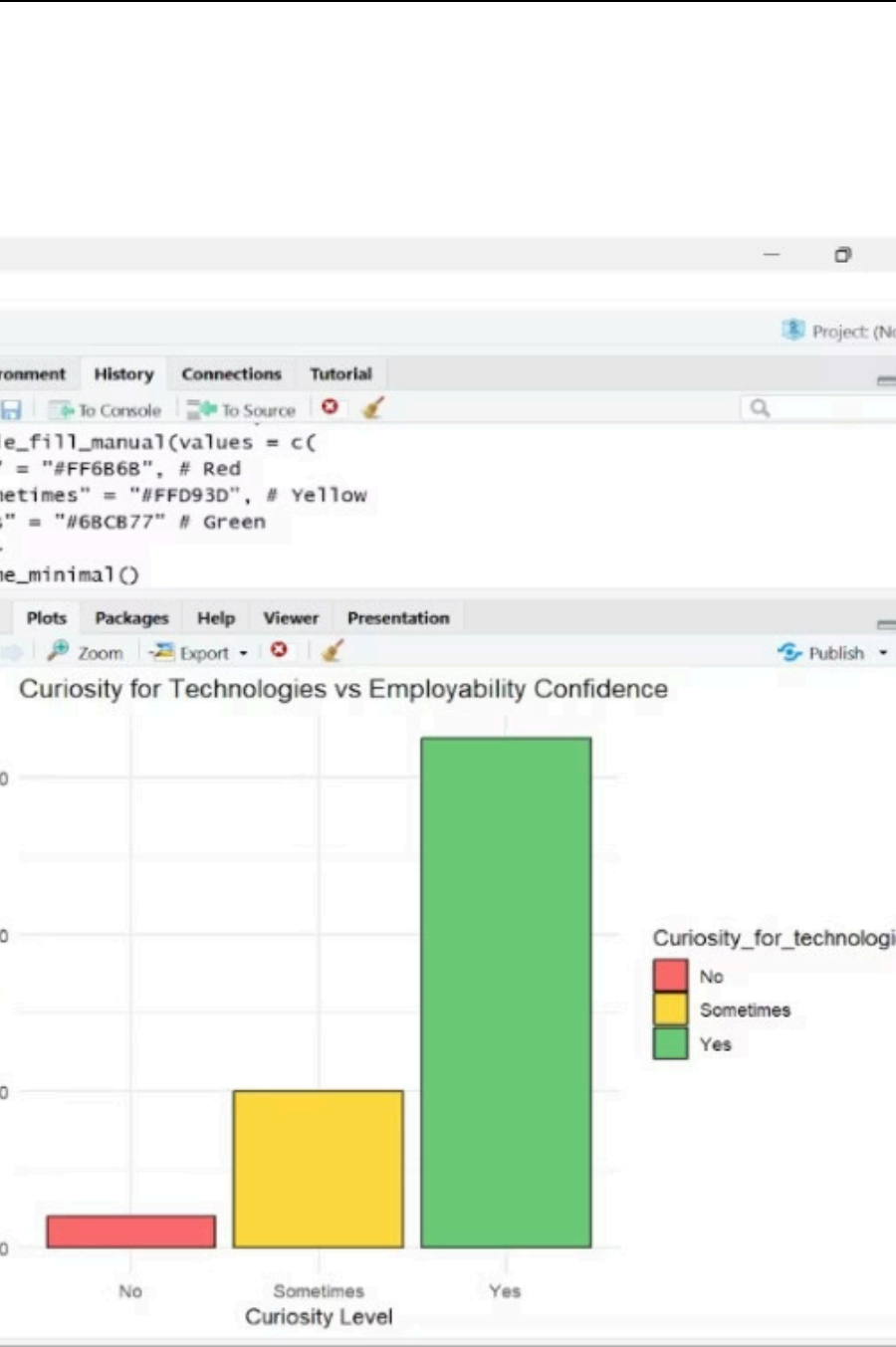
Curiosity & Confidence

Higher curiosity correlates with higher employability confidence; self-learning plays a major role

Personality Fit

Better personality match leads to higher satisfaction; mismatch increases struggle and stress

Key takeaway: Discipline, curiosity, and personality alignment are the three strongest predictors of student success and confidence in technical programs.



KEY INSIGHTS

Curiosity Fuels Satisfaction in Technical Degrees

A strong correlation exists between a student's intrinsic curiosity towards technology and their overall satisfaction and success in technical degree programs.



Ignited Curiosity

An innate desire to explore and understand new technical concepts and tools.



Deeper Engagement

Translates into proactive self-learning, experimentation, and critical thinking.



Enhanced Competence

Leads to a stronger grasp of complex subjects and improved problem-solving skills.



Higher Satisfaction

Results in a greater sense of achievement, belonging, and fulfillment with their degree.

KEY INSIGHTS

Study Consistency Elevates Satisfaction

Maintaining a consistent study schedule is a critical factor, directly influencing student performance, confidence, and overall satisfaction within demanding technical degree programs.



Consistent Routine

Regular and disciplined study schedules build a strong foundation, preventing last-minute cramming and promoting steady progress.



Deeper Comprehension

Consistent engagement allows for a more thorough understanding and retention of complex technical concepts and problem-solving methodologies.



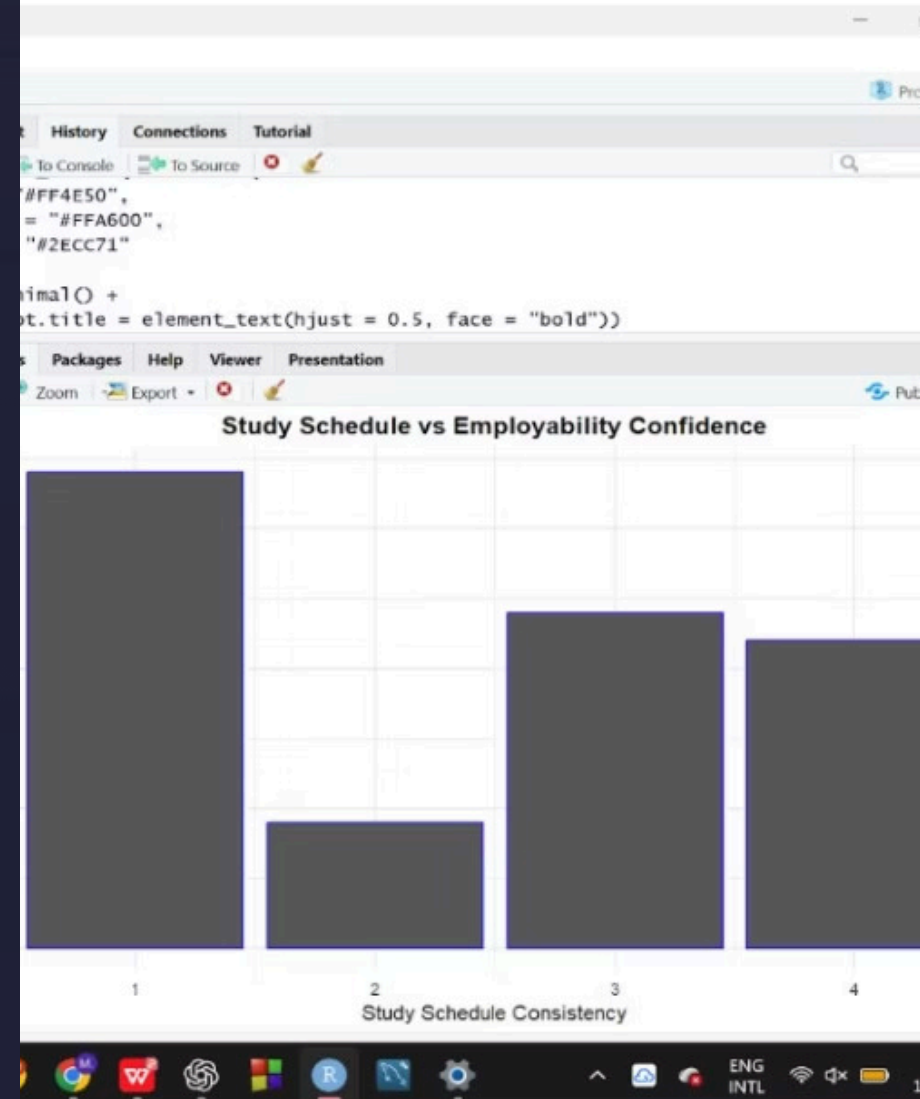
Reduced Stress

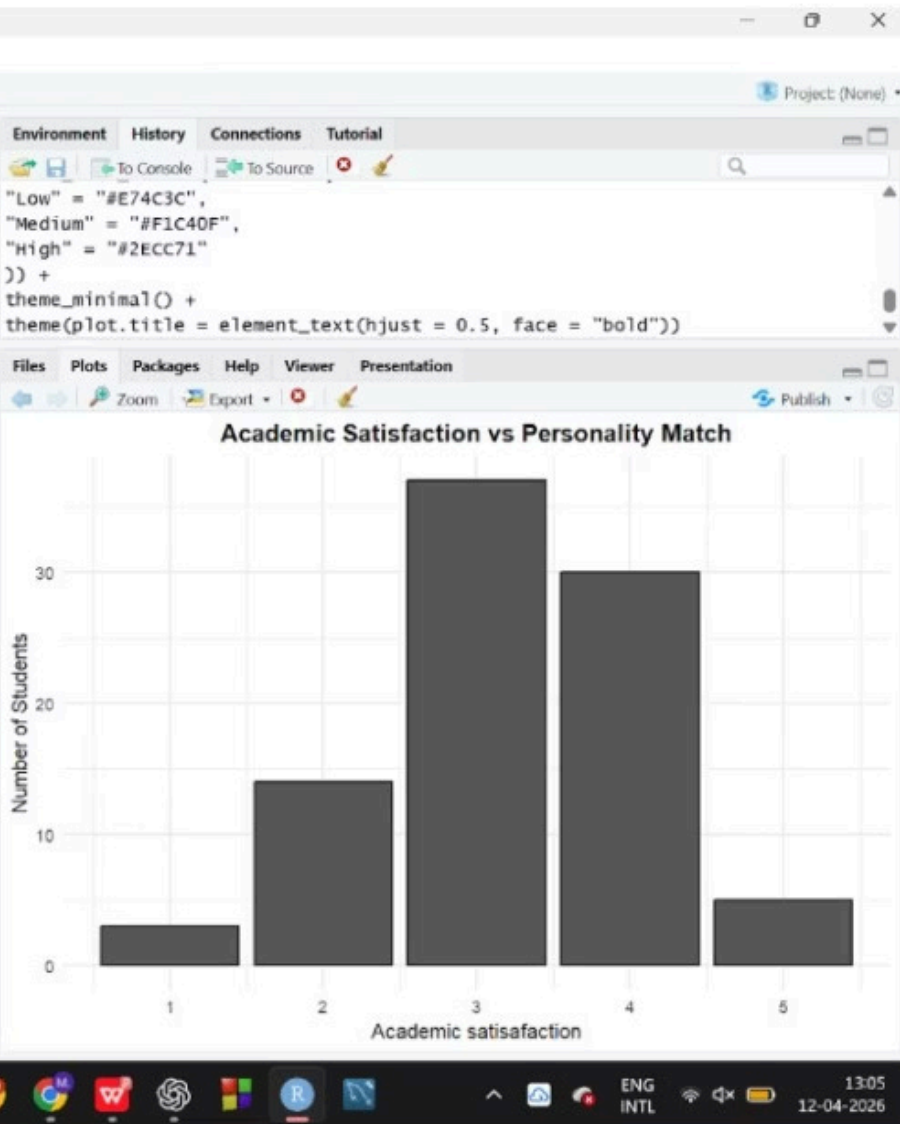
Predictable study patterns minimize academic anxiety, fostering a sense of control, preparedness, and mental well-being among students.



Higher Satisfaction

Ultimately, this leads to greater academic success, increased self-efficacy, and a more fulfilling educational experience, reinforcing positive learning loops.





KEY INSIGHTS

Personality Alignment: Driving Degree Satisfaction

When a student's innate personality traits align with the demands and environment of their technical degree program, the pathway to success and fulfillment becomes significantly clearer and more rewarding.

Self-Awareness

Students understand their unique strengths and preferences, enabling informed program choices.

Optimal Match

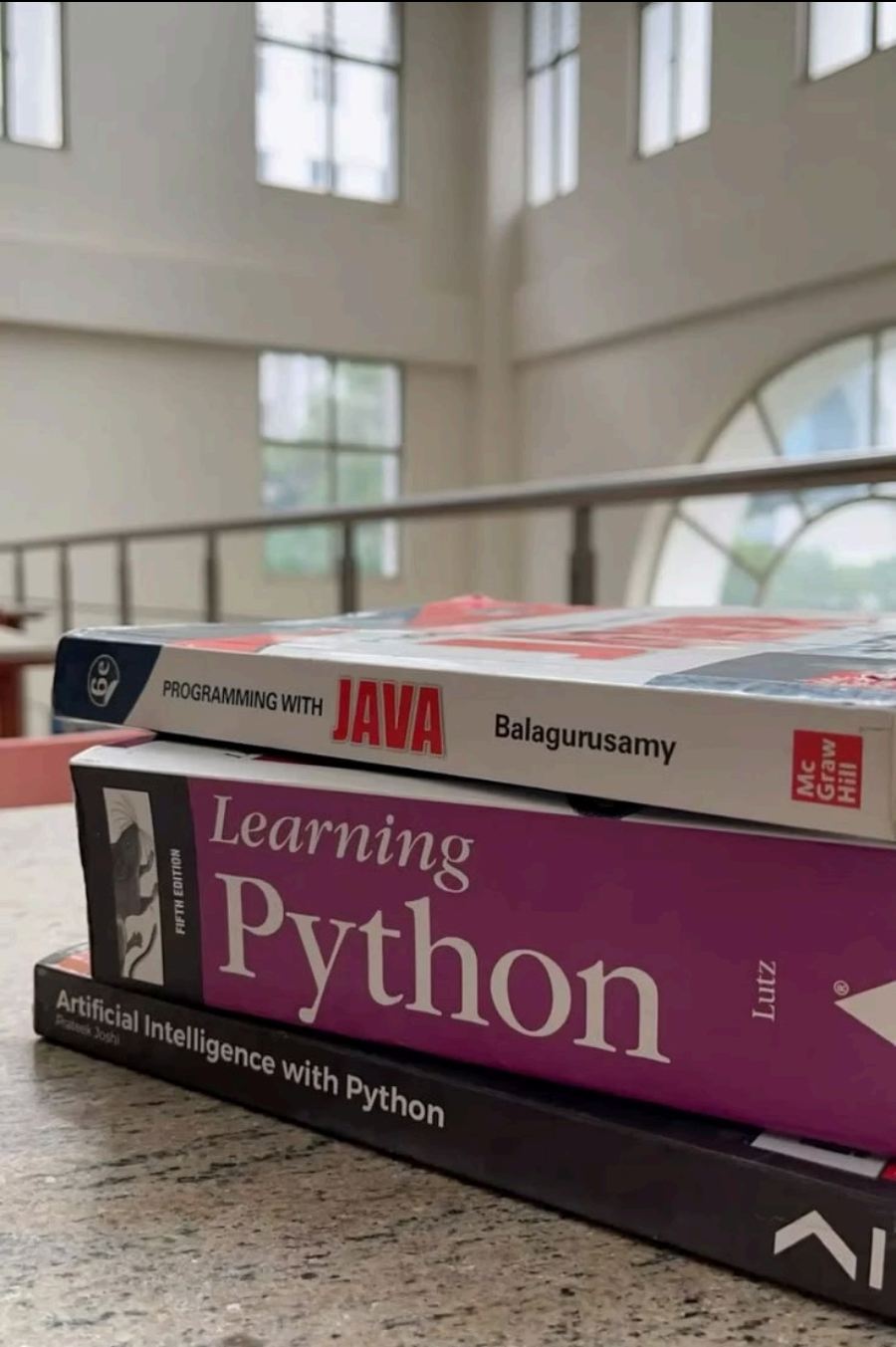
Choosing a technical field that naturally complements personal inclinations and learning styles.

Reduced Friction

Experience less struggle and greater engagement, making academic challenges feel more manageable.

Enhanced Satisfaction

Leads to higher motivation, better academic performance, and a deeper sense of accomplishment.



INSIGHTS

Key Insights

What the Data Reveals

Findings point to critical gaps in how technical education systems accommodate diverse learner personalities.

01

Personality & Learning

Personality significantly affects learning outcomes and academic performance

02

Practical Exposure

Practical, hands-on learning is essential for student engagement and skill development

03

Anxiety & Curriculum

Anxiety negatively impacts performance; current curriculum lacks flexibility to accommodate diverse learners

REFERENCES

References

Big Five Personality Traits

Goldberg, L. R. (1993). The structure of phenotypic personality traits.

MBTI Framework

Myers, I. B. & Briggs, K. C. — Myers-Briggs Type Indicator personality assessment model.

Tools & Packages

R Programming Documentation · ggplot2 visualization package · tm (Text Mining) package

Primary Data

Google Forms Survey — ~80 BCA/BTech students, 30+ questions, quantitative & qualitative responses.

Thank You

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